Cryptography

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Contents

1 Divisibility

Prop 1.1: Properties of Divisibility. *Let $a, b, c \in \mathbb{Z}$. The following properties hold:*

1. *If $a \mid 1$, then $a = \pm 1$.*
2. *If $a \mid b$ and $b \mid a$, and $a \neq 0, b \neq 0$, then $a = \pm b$.*
3. *If $a \mid b$ and $b \mid c$, then $a \mid c$ (Transitivity).*

Proof.

1. If $a \mid 1$: By definition, $\exists c \in \mathbb{Z} : 1 = ac$. Since $a, c \in \mathbb{Z}$, the only possible integer solutions are $(a = 1, c = 1)$ or $(a = -1, c = -1)$. Therefore, $a = \pm 1$.
2. If $a \mid b$ and $b \mid a$: By definition, $\exists c_1, c_2 \in \mathbb{Z} : b = ac_1$ and $a = bc_2$. Substituting the first equation into the second gives $a = (ac_1)c_2 = a(c_1 c_2)$. Since $a \neq 0$, we can divide by a to get $1 = c_1 c_2$. As $c_1, c_2 \in \mathbb{Z}$, the only solutions are $(c_1 = 1, c_2 = 1)$ or $(c_1 = -1, c_2 = -1)$.
   - If $c_1 = 1$, then $b = a \cdot 1 = a$.
   - If $c_1 = -1$, then $b = a \cdot (-1) = -a$.

   Thus, $a = \pm b$.
3. If $a \mid b$ and $b \mid c$: By definition, $\exists c_1, c_2 \in \mathbb{Z} : b = ac_1$ and $c = bc_2$. Substituting the first equation into the second gives $c = (ac_1)c_2 = a(c_1 c_2)$. Since $c_1 c_2 \in \mathbb{Z}$, this satisfies the definition of divisibility, so $a \mid c$. □

Prop 1.2: Euclidean Division. *Let $a \in \mathbb{N}$ and $b \in \mathbb{N}$ with $b > 0$. Then $\exists ! q, r \in \mathbb{Z} : a = bq + r$ with $0 \leq r < b$.*

Prop 1.3: Binary Representation from Repeated Division. *Let $m \in \mathbb{N}$, with $m < 2^B$. The repeated division-by-2 process:*

$$\begin{aligned} m &= 2q_0 + m_0 \\ q_0 &= 2q_1 + m_1 \\ q_1 &= 2q_2 + m_2 \\ &\dots \end{aligned}$$

terminates with $q_{B-1} = 0$, and $m = m_0 + 2m_1 + \cdots + 2^{B-1} m_{B-1}$.

Proof. The proof consists of two parts: demonstrating the polynomial form and proving that the process terminates.

Polynomial Form

We can show the relationship by algebraic substitution.

$$m = 2q_0 + m_0$$

Substitute $q_0 = 2q_1 + m_1$:

$$m = 2(2q_1 + m_1) + m_0 = 2^2 q_1 + 2m_1 + m_0$$

Substitute $q_1 = 2q_2 + m_2$:

$$m = 2^2(2q_2 + m_2) + 2m_1 + m_0 = 2^3 q_2 + 2^2 m_2 + 2m_1 + m_0$$

Continuing this process until the final step where $q_{B-2} = 2q_{B-1} + m_{B-1}$, we arrive at the final form: $m = m_0 + 2m_1 + 2^2 m_2 + \cdots + 2^{B-1} m_{B-1}$.

Termination

The process is guaranteed to terminate because the sequence of quotients is strictly decreasing and bounded.

- Since $m < 2^B$, the first division $m = 2q_0 + m_0$ implies $2q_0 \leq m < 2^B$, so $q_0 < 2^{B-1}$.
- Similarly, $q_0 < 2^{B-1}$ implies $2q_1 \leq q_0 < 2^{B-1}$, so $q_1 < 2^{B-2}$.

This pattern continues: $q_i < 2^{B-1-i}$. For the final quotient q_{B-1}, we have $q_{B-1} < 2^{B-1-(B-1)} = 2^0 = 1$. Since $q_{B-1} \in \mathbb{N} \cup \{0\}$, the only possibility is $q_{B-1} = 0$. The process terminates when the quotient becomes zero. $\square$

Rem: Upper Bound of a B-bit Number. *If m has B bits in its binary representation, i.e., $m = (m_{B-1} \dots m_0)_2$, then $m < 2^B$.*

Proof. The largest possible value for a B-bit number occurs when all bits are 1. This value can be calculated by summing the geometric series:

$$m \leq \sum_{i=0}^{B-1} 2^i = \frac{2^B - 1}{2 - 1} = 2^B - 1$$

Therefore, $m < 2^B$. $\square$

Cor 1.1: Range of a B-bit Number. *A natural number $m \in \mathbb{N}$ has a binary representation with B bits (where the most significant bit $m_{B-1} = 1$) $\iff$ $2^{B-1} \leq m < 2^B$.*

2 The Greatest Common Divisor (GCD)

Def 2.1: Greatest Common Divisor. *Let* $a, b \in \mathbb{Z}$*, such that* $a \neq 0$ *or* $b \neq 0$*. The set of their common divisors has a largest element* $d \in \mathbb{N}$*, called the greatest common divisor of* a *and* b*. We write* $d = \gcd(a, b)$*.*

Rem: Example of GCD. *To find* $\gcd(18, 12)$*, we can list the divisors of each number:*

- *Divisors of 12:* $\{1, 2, 3, 4, 6, 12\}$
- *Divisors of 18:* $\{1, 2, 3, 6, 9, 18\}$

The set of common divisors is $\{1, 2, 3, 6\}$*. The largest element in this set is* 6*, so* $\gcd(18, 12) = 6$*.*

While listing all divisors is feasible for small numbers, it is impractical for large integers. A far more powerful and efficient method is the Euclidean Algorithm.

3 The Euclidean Algorithm

Th 3.1: Euclidean Algorithm. *Let $a, b \in \mathbb{N} : a > 0, b > 0$ and $a \geq b$. The following algorithm computes* $\gcd(a, b)$ *in a finite number of steps:*

1. *Let $r_0 := a$ and $r_1 := b$.*
2. *Set $i = 1$.*
3. *Divide r_{i-1} by r_i to obtain a quotient q_i and remainder r_{i+1}: $r_{i-1} = r_i q_i + r_{i+1}$.*
4. *If $r_{i+1} = 0$, then $r_i = \gcd(a, b)$. The algorithm terminates.*
5. *Otherwise, increment $i \to i + 1$ and return to Step 3.*

Proof. Let the sequence of remainders be $r_0, r_1, \ldots, r_n, r_{n+1}$ where $d = r_n$ is the last non-zero remainder and $r_{n+1} = 0$. We must demonstrate two facts: that d is a common divisor of a and b, and that it is the greatest common divisor.

a. $d \mid a$ **and** $d \mid b$

We proceed by working backward from the end of the algorithm.

- The final division step is $r_{n-1} = r_n q_n + r_{n+1}$. Since $r_{n+1} = 0$ and $d = r_n$, this equation becomes $r_{n-1} = dq_n$. This demonstrates $d \mid r_{n-1}$.
- The preceding step is $r_{n-2} = r_{n-1}q_{n-1} + r_n$. Since $d \mid r_n$ and we have shown $d \mid r_{n-1}$, it follows that d must divide their linear combination, $d \mid r_{n-2}$.

Continuing this backward induction process, we establish that d divides every preceding remainder. This concludes that $d \mid r_1$ (where $r_1 = b$) and $d \mid r_0$ (where $r_0 = a$). Thus, d is a common divisor.

b. d **is the Greatest Common Divisor**

Let $c \in \mathbb{Z}$ be any common divisor of a and b. We work forward down the chain of remainders.

- Since $c \mid a$ (r_0) and $c \mid b$ (r_1), and $r_2 = r_0 - r_1 q_1$, it must be that $c \mid r_2$.
- Since $c \mid r_1$ and $c \mid r_2$, and $r_3 = r_1 - r_2 q_2$, it must be that $c \mid r_3$.

Continuing this process, we eventually find that c must divide the last non-zero remainder, r_n, so $c \mid d$. Since $c \mid d$, it is a consequence of divisibility that $c \leq d$. This holds $\forall c$ common divisor, proving that d is the greatest common divisor. □

Prop 3.1: Extended Euclidean Algorithm. *Let $a, b \in \mathbb{N}$ such that $a \neq 0$ or $b \neq 0$. If $d = \gcd(a, b)$, then $\exists u, v \in \mathbb{Z} : d = au + bv$.*

Prop 3.2: Solvability Condition. *The linear Diophantine equation $au + bv = c$ has integer solutions $u, v \iff \gcd(a, b) \mid c$.*

Proof.

(⇒) Assume integer solutions u, v exist. Let $d = \gcd(a, b)$. By definition, $d \mid a$ and $d \mid b$. $\implies d \mid (au + bv)$, $\implies d \mid c$ because $au + bv = c$.

(⇐) Assume $d = \gcd(a, b)$ and $d \mid c$. Then $\exists k \in \mathbb{Z} : c = kd$. From the Extended Euclidean Algorithm, we know $\exists u_0, v_0 \in \mathbb{Z} : au_0 + bv_0 = d$. Multiplying by k, we get $a(ku_0) + b(kv_0) = kd = c$. Thus, a solution exists with $u = ku_0$ and $v = kv_0$. □

Prop 3.3: Form of All Solutions. *If (u_p, v_p) is a particular solution to $au + bv = c$ and $d = \gcd(a, b)$, let $a' = a/d$ and $b' = b/d$. Then $\forall (u, v)$ integer solutions, $u = u_p + kb'$ and $v = v_p - ka'$ for any $k \in \mathbb{Z}$.*

Proof. Assume (u', v') is an arbitrary integer solution, so $au' + bv' = c$. We also have $au_p + bv_p = c$. Subtracting the two equations gives $a(u' - u_p) + b(v' - v_p) = 0$, so $a(u' - u_p) = b(v_p - v')$. Dividing by d: $a'(u' - u_p) = b'(v_p - v')$. Since $\gcd(a', b') = 1$, and $a' \mid b'(v_p - v')$, it must be that $a' \mid (v_p - v')$. $\implies \exists k \in \mathbb{Z} : v_p - v' = ka'$, which gives $v' = v_p - ka'$. Substituting this back yields $a'(u' - u_p) = b'(ka')$, which simplifies to $u' - u_p = kb'$, or $u' = u_p + kb'$. This demonstrates that any other solution must be of the specified form. □

4 Coprime numbers

Prop 4.1: Characterisation of Coprime Numbers. *Integers $a, b \in \mathbb{Z}$ are relatively prime $\iff \exists u, v \in \mathbb{Z} : au + bv = 1$.*

Proof.

(⇒) This is a direct consequence of the Extended Euclidean Algorithm. If $\gcd(a, b) = 1$, then the algorithm guarantees $\exists u, v \in \mathbb{Z} : au + bv = 1$.

(⇐) Conversely, assume $\exists u, v \in \mathbb{Z} : au + bv = 1$. Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, it must divide their linear combination $au + bv$. Therefore, $d \mid 1$. The only $d \in \mathbb{N}$ so that $d \mid 1$ is $d = 1$. □

Cor 4.1. *Let p be a prime number and $m \in \mathbb{Z}$. If $p \nmid m$, then* $\gcd(p, m) = 1$.

Proof. The positive divisors of a prime number p are only $\{1, p\}$. If $p \nmid m$, then p is not a common divisor. Therefore, the only common divisor of p and m is 1, which means $\gcd(p, m) = 1$. □

Prop 4.2: Prime Divisor Property. *Let p be a prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Assume $p \mid ab$. If $p \mid a$, the proposition holds. If $p \nmid a$, then by the preceding corollary, $\gcd(p, a) = 1$. By the characterization of coprime numbers, $\exists u, v \in \mathbb{Z} : pu + av = 1$. Multiplying this equation by b gives $pub + abv = b$.

Since $p \mid pub$ and we are given $p \mid ab$, which implies $p \mid abv$. Since p divides both terms on the left side, it must divide their sum. Therefore, $p \mid (pub + abv) \to p \mid b$. □

Th 4.1: Fundamental Theorem of Arithmetic. *Let $a \in \mathbb{N}$ so that $a \geq 2$. Then a can be factored as a product of primes, $a = p_1^{e_1} p_2^{e_2} \cdots p_m^{e_m}$, where p_i are distinct primes, and $\forall i \in \{1, \ldots, m\}, e_i \geq 1$. The factorization into primes is unique modulo rearrangements of the order of the factors.*

Rem. *The exponent $e_i = mult_{p_i}(a)$ is the multiplicity of p_i in a. A generalized form of the prime factorization is given by:*

$$a = \prod_{p\ prime} p^{mult_p(a)}$$

where $mult_p(a) := 0$ if p is not a prime factor of a.

5 Classes Modulo m

Prop 5.1. *$\forall a \in \mathbb{Z}$ (the dividend) and $\forall m \in \mathbb{N}$ so that $m > 1$ (the modulus), $\exists! q, r \in \mathbb{Z} : a = mq + r$, with $0 \leq r < m$.*

Th 5.1: Existence of a Unique Remainder. *$\forall a \in \mathbb{Z}$ and $\forall m \in \mathbb{N} : m > 1$, $\exists! r \in \{0, 1, \dots, m-1\} : a \equiv r \pmod{m}$.*

Proof. By Proposition 5.1 (Euclidean Division), we have $\exists! q, r \in \mathbb{Z} : a = mq + r$, where $0 \leq r < m$. Rearranging this equation gives $a - r = mq$. By the formal definition of divisibility, $m \mid (a - r)$. By the formal definition of congruence, this directly implies $a \equiv r \pmod{m}$. The uniqueness of r is guaranteed by the Euclidean Division Theorem. □

Th 5.2: Congruence is an Equivalence Relation. *The relation of congruence modulo m (denoted by $\equiv \pmod{m}$) is an equivalence relation on the set of integers $\mathbb{Z}$.*

Proof. We must demonstrate that congruence modulo m satisfies reflexivity, symmetry, and transitivity.

- **Reflexivity:** $\forall a \in \mathbb{Z}$, $a - a = 0$. Since $m \mid 0$, $a \equiv a \pmod{m}$. The relation is reflexive.
- **Symmetry:** Assume $a \equiv b \pmod{m}$. $\implies m \mid (a - b)$, so $\exists k \in \mathbb{Z} : a - b = km$. Multiplying by -1 yields $b - a = (-k)m$. Since $-k \in \mathbb{Z}$, $\implies m \mid (b - a)$, so $b \equiv a \pmod{m}$. The relation is symmetric.
- **Transitivity:** Assume $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. $\implies \exists k_1, k_2 \in \mathbb{Z} : a - b = k_1 m$ and $b - c = k_2 m$. Adding the equations:

$$(a - b) + (b - c) = k_1 m + k_2 m$$

$$a - c = (k_1 + k_2) m$$

Since $k_1 + k_2 \in \mathbb{Z}$, $\implies m \mid (a - c)$, so $a \equiv c \pmod{m}$. The relation is transitive. □

Prop 5.2: Equivalence Classes Form a Partition. *Let $\sim$ be an equivalence relation on a set X. The set of equivalence classes, $X/\sim$, forms a partition of X.*

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6 Invertibility Modulo m

Prop 6.1: Compatibility. *Let $a_1, a_2, b_1, b_2 \in \mathbb{Z}$. If $a_1 \equiv a_2 \pmod{m}$ and $b_1 \equiv b_2 \pmod{m}$, then:*

1. $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$
2. $a_1 b_1 \equiv a_2 b_2 \pmod{m}$

Proof.

- **Addition:** Given $a_2 \equiv a_1 \pmod{m}$ and $b_2 \equiv b_1 \pmod{m}$, $\implies \exists k_1, k_2 \in \mathbb{Z} : a_2 - a_1 = k_1 m$ and $b_2 - b_1 = k_2 m$. Adding the equations yields $(a_2 - a_1) + (b_2 - b_1) = (k_1 + k_2)m$. Rearranging the terms: $(a_2 + b_2) - (a_1 + b_1) = (k_1 + k_2)m$. Since $k_1 + k_2 \in \mathbb{Z}$, this shows $m \mid ((a_2 + b_2) - (a_1 + b_1))$, so $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$.
- **Multiplication:** We express $a_2 = a_1 + k_1 m$ and $b_2 = b_1 + k_2 m$. Multiplying them gives:

  $$a_2 b_2 = (a_1 + k_1 m)(b_1 + k_2 m) = a_1 b_1 + a_1 k_2 m + b_1 k_1 m + k_1 k_2 m^2$$

  Factoring out m: $a_2 b_2 - a_1 b_1 = m(a_1 k_2 + b_1 k_1 + k_1 k_2 m)$. Let $K = a_1 k_2 + b_1 k_1 + k_1 k_2 m$. Since $K \in \mathbb{Z}$, $m \mid (a_2 b_2 - a_1 b_1)$, so $a_1 b_1 \equiv a_2 b_2 \pmod{m}$. $\square$

Prop 6.2. *The set of integers modulo m under addition, $(\mathbb{Z}/m\mathbb{Z}, +)$, forms a **commutative group**.*

Prop 6.3. *When considering both addition and multiplication, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$ forms a **commutative ring**.*

Th 6.1. *If an inverse exists, it is unique.*

Proof. Assume an element $[a]$ has two inverses, $[a']$ and $[a'']$. By definition, $[a] \cdot [a'] = [1]$ and $[a] \cdot [a''] = [1]$. Since multiplication is associative and $[1]$ is the identity element, we can write:

$$[a'] = [a'] \cdot [1] = [a'] \cdot ([a] \cdot [a''])$$

$$[a'] = ([a'] \cdot [a]) \cdot [a''] = [1] \cdot [a''] = [a'']$$

Therefore, $[a'] = [a'']$, $\implies$ the inverse is unique. $\square$

Th 6.2: Characterization of Invertibility. *An element $[a] \in \mathbb{Z}/m\mathbb{Z}$ is invertible $\iff$ $\gcd(a, m) = 1$.*

Proof. An element $[a]$ is invertible $\iff$ $\exists u \in \mathbb{Z} : au \equiv 1 \pmod m$. This congruence is equivalent to the linear Diophantine equation $au - 1 = -mv$ for some $v \in \mathbb{Z}$, which can be rewritten as $au + mv = 1$. By Bézout's Identity, this equation has an integer solution $\iff$ $\gcd(a, m) \mid 1$. Since $\gcd(a, m) \in \mathbb{N}$, this condition is satisfied $\iff$ $\gcd(a, m) = 1$. □

Prop 6.4. *The set of units under multiplication, $(\mathbb{Z}/m\mathbb{Z})^*$, forms a **commutative group** (the group of units modulo m).*

Prop 6.5. *$\forall p$ prime number, the commutative ring $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a **field**, commonly denoted $\mathbb{F}_p$.*

Cor 6.1. *Let $p \in \mathbb{N}$ be a prime so that $p > 2$. The equation $x^2 \equiv 1 \pmod p$ in $\mathbb{F}_p$ has exactly two distinct solutions: $x \equiv 1 \pmod p$ and $x \equiv p - 1 \pmod p$ (i.e., $x \equiv -1 \pmod p$).*

Proof. The equation $x^2 \equiv 1 \pmod p$ is equivalent to $p \mid (x^2 - 1)$, or $p \mid (x - 1)(x + 1)$. Since p is prime, by the Prime Divisor Property:

- Either $p \mid (x - 1)$, which implies $x \equiv 1 \pmod p$.
- Or $p \mid (x + 1)$, which implies $x \equiv -1 \equiv p - 1 \pmod p$.

These two solutions are distinct because $1 \not\equiv -1 \pmod p$ for $p > 2$. □

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7 Solving Linear Congruences

Prop 7.1: Solvability Condition. *The linear congruence* $ax \equiv b \pmod{m}$ *has at least one integer solution for* $x \in \mathbb{Z} \iff d = \gcd(a, m) \mid b$.

Proof. The congruence $ax \equiv b \pmod{m}$ is equivalent to the linear Diophantine equation $ax + mv = b$ for some $v \in \mathbb{Z}$. This equation is solvable for $x, v \in \mathbb{Z} \iff$ the greatest common divisor of the coefficients, $d = \gcd(a, m)$, divides the constant term b. This follows directly from the Solvability Condition for Diophantine Equations (Bézout's identity). □

Prop 7.2: Number of Solutions. *If* $d = \gcd(a, m) \mid b$*, the congruence* $ax \equiv b \pmod{m}$ *has exactly* d *distinct solutions modulo* m.

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8 Chinese Remainder Theorem (CRT)

Th 8.1: Chinese Remainder Theorem. *Let $m_1, m_2, \ldots, m_k \in \mathbb{N}$ be pairwise relatively prime integers (i.e., $\forall i \neq j, \gcd(m_i, m_j) = 1$). Let $a_1, a_2, \ldots, a_k \in \mathbb{Z}$ be arbitrary integers. The system of congruences:*

$$x \equiv a_i \pmod{m_i} \quad \forall i \in \{1, \ldots, k\}$$

has a solution, and the complete set of solutions is congruent modulo the product $M = m_1 \cdot m_2 \cdot \cdots \cdot m_k$.

Proof of the Chinese Remainder Theorem. The proof is divided into two parts: Existence (constructive) and Uniqueness (based on coprimality).

Proof of Existence

We define the solution constructively as $x = \sum_{j=1}^{k} a_j v_j N_j$. For this construction, let $M = \prod_{i=1}^{k} m_i$. We define $N_j = M/m_j$. Since the m_i are pairwise coprime, $\gcd(N_j, m_j) = 1$. $\implies \exists v_j \in \mathbb{Z} : v_j N_j \equiv 1 \pmod{m_j}$ (the multiplicative inverse of N_j modulo m_j). We must show that $x \equiv a_i \pmod{m_i}$ for an arbitrary index $i \in \{1, \ldots, k\}$.

The core argument is that $\forall j \neq i$, the term N_j contains m_i as a factor (since m_i is a factor of M but not m_j).

$$\forall j \neq i, \quad m_i \mid N_j \quad \implies a_j v_j N_j \equiv 0 \pmod{m_i}$$

Thus, when x is taken modulo m_i, all terms in the sum vanish except the i-th term ($j = i$):

$$x \equiv a_i v_i N_i + \sum_{j \neq i} (0) \pmod{m_i}$$

$$x \equiv a_i v_i N_i \pmod{m_i}$$

Since $v_i N_i \equiv 1 \pmod{m_i}$ by construction, the proof concludes:

$$x \equiv a_i(1) \equiv a_i \pmod{m_i}$$

This confirms the existence of a solution x that satisfies the system.

Proof of Uniqueness

1. **Assume Two Solutions:** Let x and x' both solve the system.
2. **Relationship:** $\forall i \in \{1, \ldots, k\}$, $x \equiv a_i \pmod{m_i}$ and $x' \equiv a_i \pmod{m_i}$. $\implies x - x' \equiv 0 \pmod{m_i}$. This means that $\forall i$, $m_i \mid (x - x')$.
3. **Use Co-prime Condition:** Since the moduli $m_1, \ldots, m_k$ are pairwise co-prime and each divides the difference $(x - x')$, their product $M = \prod_{i=1}^{k} m_i$ must also divide the difference.
4. **Conclusion:** $M \mid (x - x')$ means $x \equiv x' \pmod{M}$.

This confirms that the solution is unique modulo M. □

9 Euler's ϕ Function

Th 9.1: Totient of a Prime Number. *The Totient of a Prime Number: If $m \in \mathbb{N}$ so that m is prime, then $\varphi(m) = m - 1$.*

Th 9.2: Primality Test via the Totient Function. *A Primality Test via the Totient Function: $\forall m \in \mathbb{N} : m \geq 2$, $\varphi(m) = m - 1 \iff m$ is a prime number.*

Proof.

($\Rightarrow$) **("Only if" direction)**: We prove the contrapositive. Assume m is composite (not prime). $\implies \exists p \in \mathbb{N} : 1 < p < m$ and $p \mid m$. Since $p \mid m$, $\gcd(p, m) = p > 1$. Thus, p is not coprime to m. Since $p \in \{1, \ldots, m-1\}$ is excluded from the count, the total count $\varphi(m)$ must be strictly less than $m - 1$. More precisely, $\varphi(m) \leq m - 2$. This shows that the equality $\varphi(m) = m - 1$ holds $\implies m$ is prime.

($\Leftarrow$) **("If" direction)**: Assume m is prime. Then $\forall a \in \{1, \ldots, m-1\}$, $\gcd(a, m) = 1$. Since there are $m - 1$ such integers, $\varphi(m) = m - 1$. □

Prop 9.1: Totient of a Prime Power. *$\forall p$ prime and $\forall \alpha \in \mathbb{N} : \alpha \geq 1$:*

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$$

Proof. The totient function $\varphi(p^\alpha)$ counts the number of integers $a \in \{1, \ldots, p^\alpha\}$ so that $\gcd(a, p^\alpha) = 1$. The integers that are **not** coprime to p^α are those a for which $p \mid a$. We count these integers (the complement). These non-coprime integers are the multiples of p in the range $[1, p^\alpha]$, i.e., of the form $k \cdot p$ where $k \in \mathbb{N}$. The condition $1 \leq k \cdot p \leq p^\alpha$ implies $1 \leq k \leq p^{\alpha-1}$. Thus, there are exactly $p^{\alpha-1}$ multiples of p in the specified range. Subtracting this count from the total number of integers, p^α, yields:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p - 1)$$

□

Th 9.3: Multiplicativity of Euler's Totient Function. *Multiplicativity of Euler's Totient Function: If $m, n \in \mathbb{N}$ are coprime integers ($\gcd(m, n) = 1$), then $\varphi(mn) = \varphi(m)\varphi(n)$.*

10 The fast powering algorithm

Th 10.1: Solvability of a System of Two Linear Congruences. *Let m and n be positive integers, and let a and b be any integers. The system of linear congruences:*

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

has a solution if and only if $\gcd(m, n) \mid (a - b)$.

1: Constructive Proof by Cases (Alessandra's Method). We prove the biconditional nature by demonstrating necessity ($\implies$) and sufficiency ($\impliedby$) separately.

1. Necessity ($\implies$): We assume a solution exists $\to \exists x_0 \in \mathbb{Z} : x_0 \equiv a \pmod{m}$ and $x_0 \equiv b \pmod{n}$. This implies:

$$x_0 - a = mk \quad \text{for some } k \in \mathbb{Z}$$

$$x_0 - b = nl \quad \text{for some } l \in \mathbb{Z}$$

Thus, $x_0 = a+mk$ and $x_0 = b+nl$. Equating the two expressions for x_0 yields $a+mk = b+nl$. Rearranging the terms: $a - b = nl - mk$. Let $d = \gcd(m, n)$. By definition, $d \mid m$ and $d \mid n$. Therefore, d must divide any integer linear combination of m and n, $\forall k, l \in \mathbb{Z}$. Since $a - b = nl - mk$, we conclude that $d \mid (a - b)$.

2. Sufficiency ($\impliedby$): We assume $\gcd(m, n) \mid (a - b)$. Let $d = \gcd(m, n)$. Since $d \mid (a - b)$, we have $a - b = kd$ for some integer $k \in \mathbb{Z}$. By Bezout's identity, $\exists u, v \in \mathbb{Z} : d = mu + nv$. Substituting the expression for d: $a - b = k(mu + nv) \to a - b = kmu + knv$. Rearranging the equation: $a - kmu = b + knv$. Let $x_0 = a - kmu$. By construction: $x_0 = a - kmu \to x_0 - a = (-ku)m$. Since $-ku \in \mathbb{Z}$, $x_0 \equiv a \pmod{m}$. Also, $x_0 = b + knv \to x_0 - b = (kv)n$. Since $kv \in \mathbb{Z}$, $x_0 \equiv b \pmod{n}$. Since x_0 satisfies both congruences, a solution exists. □

2: Direct Proof via Reduction (Professor's Method). The first congruence $x \equiv a \pmod{m}$ can be rewritten as $x = a+km$ for some integer $k \in \mathbb{Z}$. Substituting this into the second congruence yields:

$$a + km \equiv b \pmod{n}$$

Rearranging the terms, we obtain a single linear congruence for the unknown k:

$$km \equiv b - a \pmod{n}$$

A solution for this linear congruence exists if and only if $\gcd(m, n) \mid (b-a)$. Since $\gcd(m, n) = \gcd(m, n)$ and $\gcd(m, n) \mid (b - a)$ is equivalent to $\gcd(m, n) \mid (a - b)$, the original system has a solution if and only if the condition holds. □

Prop 10.1: General Solvability Condition. *A system of congruences $x \equiv a_i \pmod{m_i}$ for $i \in \{1, 2, \dots, n\}$ has a solution if and only if for every pair of indices $i \neq j$, the condition $\gcd(m_i, m_j) \mid (a_j - a_i)$ holds.*

A Method to Compute $g^A \pmod N$

Alg 10.1: Fast Powering Algorithm. *Let $g, A, N \in \mathbb{N}$ such that $g \geq 1$, $A \geq 1$, and $N \geq 1$.*

Let $A = (A_r A_{r-1} \ldots A_0)_2$ be the binary representation of A, so that $A = A_0 + 2A_1 + 2^2 A_2 + \cdots + 2^r A_r$, where $A_i \in \{0, 1\}$ for $0 \leq i \leq r$, and $A_r = 1$.

1. ***Compute the powers $g^{2^i} \pmod N$ for $0 \leq i \leq r$ by successive squaring:***

$$\begin{aligned} a_0 &\equiv g^{2^0} \equiv g \pmod N \\ a_1 &\equiv a_0^2 \equiv g^{2^1} \pmod N \\ a_2 &\equiv a_1^2 \equiv g^{2^2} \pmod N \\ &\vdots \\ a_r &\equiv a_{r-1}^2 \equiv g^{2^r} \pmod N \end{aligned}$$

   This step requires r multiplications. The next one is the square of the previous one.

2. ***Compute $g^A \pmod N$ using the formula $g^A = g^{A_0 + 2A_1 + 2^2 A_2 + \cdots + 2^r A_r}$:***

$$\begin{aligned} g^A &\equiv g^{A_0} \cdot (g^2)^{A_1} \cdot (g^{2^2})^{A_2} \cdots (g^{2^r})^{A_r} \pmod N \\ &\equiv a_0^{A_0} \cdot a_1^{A_1} \cdot a_2^{A_2} \cdots a_r^{A_r} \pmod N \end{aligned}$$

   This step requires at most r multiplications (since $A_i \in \{0, 1\}$).

In total (once the binary form of A is known), we need $2r \approx 2\log_2 A$ steps instead of A (naive method)!

Number of Operations

Rem: Complexity of the Fast Powering Algorithm. *If $2^r \leq A < 2^{r+1}$, then r is the number of bits in the binary representation of A ($\approx \log_2 A$).*

1. *It requires at most r divisions to write A in binary ($\sim r$ bits).*
2. *To compute g^A, where $A = A_0 + 2A_1 + 2^2 A_2 + \cdots + 2^r A_r$, we need:*
   - *r multiplications for the successive squaring (Step 1).*
   - *At most r multiplications for the final product (Step 2).*

   Thus, we need $\sim 2r$ multiplications to compute $g^A \pmod N$.

   Since $2^r \leq A$, it follows that $r \leq \log_2(A)$. Therefore, the total number of multiplications is $\leq 2r \leq 2\log_2(A)$.

Prop 10.2: Efficiency Comparison. *If $A \approx 2^{1000} \approx 10^{300}$, then $\log_2(A) \approx 1000$. The Fast Powering Algorithm requires $\sim 2\log_2(A) \approx 2000$ multiplications.*

In contrast, the Naive Method requires $A \approx 10^{300}$ multiplications. The Fast Powering Algorithm is dramatically more efficient.

Rem: Problem Statement. *The algorithm is designed to efficiently calculate the value of $g^A \pmod N$, given integers $g > 1, A > 1$, and $N > 1$. The challenge arises when the exponent A is very large, $\forall A \in \mathbb{N} : A \gg 1$. The naive method requires $\mathcal{O}(A)$ multiplications, which is computationally infeasible for large A.*

Rem: Method Principle. *The Fast Powering Algorithm leverages the binary representation of the exponent A. Let the binary expansion be $A = \sum_{i=0}^{r} A_i 2^i$, where $A_i \in \{0, 1\}$. The exponentiation property allows us to write:*

$$g^A = g^{\sum_{i=0}^{r} A_i 2^i} = \prod_{i=0}^{r} (g^{2^i})^{A_i} \pmod N$$

*The core idea is **Successive Squaring**: the terms $g^{2^i} \pmod N$ are computed efficiently by starting with g^1 and repeatedly squaring the previous result, $\forall i \in \{1, \dots, r\} : g^{2^i} = (g^{2^{i-1}})^2 \pmod N$. The final result is the product of only those terms $g^{2^i} \pmod N$ for which the corresponding binary coefficient $A_i = 1$.*

Rem: Complexity Analysis. *The Fast Powering Algorithm significantly reduces complexity from $\mathcal{O}(A)$ (naive method) to $\mathcal{O}(\log_2(A))$. This is due to the three phases:*

- *Binary Conversion: $\sim \log_2(A)$ divisions.*
- *Successive Squaring: $\sim \log_2(A)$ squarings.*
- *Final Product: at most $\sim \log_2(A)$ multiplications.*

The total number of operations $\sim 3\log_2(A)$, providing immense efficiency gain, $\forall A \in \mathbb{N} : A \approx 2^{1000} \to$ Operations ≈ 3000.

Th 10.2: Fermat's Little Theorem - Primary Form. *Let p be a prime number. For any integer $a \in \mathbb{Z}$ so that p does not divide a, we have:*

$$a^{p-1} \equiv 1 \pmod p$$

Th 10.3: Fermat's Little Theorem - Alternative Form. *Let p be a prime number. For any integer $a \in \mathbb{Z}$, we have:*

$$a^p \equiv a \pmod{p}$$

Rem: Conditions for Application. *The theorem is generally false for composite moduli. The primary condition is that the modulus p must be a prime number.*

Th 10.4: Fermat's Little Theorem. *Let p be a prime. If $a \in (\mathbb{Z}/p\mathbb{Z})^*$, then $a^{p-1} = 1$. Equivalently, if p is a prime and a is an integer so that p does not divide a, then $a^{p-1} \equiv 1 \pmod{p}$.*

Proof. The proof employs a technique involving the set of invertible elements modulo p.

Step 1: Construct the Set Let $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Consider the set S formed by multiplying a by each of the $p-1$ invertible elements of $\mathbb{Z}/p\mathbb{Z}$, which are $I = \{1, 2, \ldots, p-1\}$.

$$S = \{a \cdot i \pmod{p} \mid \forall i \in I\}$$

Step 2: Prove Non-Zero Property We claim that $a \cdot i \not\equiv 0 \pmod{p}$ for all $i \in I$. Assume for contradiction that $a \cdot i \equiv 0 \pmod{p}$. This implies $p \mid (a \cdot i)$. Since p is prime, $p \mid a$ or $p \mid i$. By premise, $a \in (\mathbb{Z}/p\mathbb{Z})^*$, so $p \nmid a$. Since $1 \leq i \leq p-1$, $\gcd(i,p) = 1$, so $p \nmid i$. This is a contradiction, so $\forall i \in I$, $a \cdot i \not\equiv 0 \pmod{p}$.

Step 3: Prove Distinctness We claim that all $p-1$ elements in S are distinct. Assume for contradiction that $\exists i, j \in I$ so that $i \neq j$ and $a \cdot i \equiv a \cdot j \pmod{p}$. This implies $a \cdot (i-j) \equiv 0 \pmod{p}$. Since $p \nmid a$, we must have $p \mid (i-j)$. However, since $i, j \in \{1, \ldots, p-1\}$ and $i \neq j$, the difference $i-j$ satisfies $0 < |i-j| < p$. Such an integer cannot be a multiple of p. This contradiction proves that all elements in S are distinct.

Step 4: Equate Sets Since S contains $p-1$ distinct, non-zero elements modulo p, and the set of all non-zero elements is $(\mathbb{Z}/p\mathbb{Z})^* = I = \{1, 2, \ldots, p-1\}$, the two sets must be identical, $\forall i \in I : S = I$.

Step 5: The Trick (Multiplication) The product of the elements in S must equal the product of the elements in I:

$$\prod_{i=1}^{p-1} (a \cdot i) \equiv \prod_{i=1}^{p-1} i \pmod{p}$$

Step 6: Simplify and Conclude Factoring the left-hand side yields:

$$a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}$$

Since p is prime, $(p-1)!$ is invertible modulo p (by Wilson's Theorem, or simply because $\forall i \in \{1, \ldots, p-1\}$, $\gcd(i,p) = 1$). We can safely cancel $(p-1)!$ from both sides:

$$a^{p-1} \equiv 1 \pmod{p}$$

$\square$

Cor 10.1: Modular Exponentiation. *Let p be a prime, and $a \in \mathbb{Z}$. If $\forall x, y \in \mathbb{N} \setminus \{0\}$ so that $x \equiv y \pmod{p-1}$, then $a^x \equiv a^y \pmod{p}$.*

Proof. We consider two distinct cases for the integer a.

Case 1: p divides a If $p \mid a$, then $a \equiv 0 \pmod{p}$. Since $x \geq 1$ and $y \geq 1$, we have $a^x \equiv 0 \pmod{p}$ and $a^y \equiv 0 \pmod{p}$. Thus, $a^x \equiv a^y \pmod{p}$ holds trivially.

Case 2: p does not divide a The congruence $x \equiv y \pmod{p-1}$ means $y = x + k(p-1)$ for some $k \in \mathbb{Z}$. Substituting this into a^y:

$$a^y = a^{x+k(p-1)} = a^x \cdot a^{k(p-1)} = a^x \cdot (a^{p-1})^k$$

By Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod{p}$. Therefore:

$$a^y \equiv a^x \cdot (1)^k \pmod{p} \to a^y \equiv a^x \pmod{p}$$

□

Cor 10.2: Finding Modular Inverses. *Let p be a prime and $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Then the inverse of a modulo p is a^{p-2}.*

Proof. We must show that $a \cdot a^{p-2} \equiv 1 \pmod{p}$.

$$a \cdot a^{p-2} = a^{1+p-2} = a^{p-1}$$

By Fermat's Little Theorem, $\forall a \in \mathbb{Z}$ so that $p \nmid a$, we have $a^{p-1} \equiv 1 \pmod{p}$. Therefore:

$$a \cdot a^{p-2} \equiv 1 \pmod{p}$$

This confirms that a^{p-2} is the multiplicative inverse of a modulo p. □

Cor 10.3: Fermat Primality Test. *Let p be an integer so that $p \geq 2$, and let a be an integer so that p does not divide a. If $a^{p-1} \not\equiv 1 \pmod{p}$, then p is not a prime number.*

Proof. This statement is the contrapositive of Fermat's Little Theorem: $(\neg\text{THEN}) \implies (\neg\text{IF})$. The original theorem states: IF p is prime, THEN $a^{p-1} \equiv 1 \pmod{p}$. The contrapositive is: IF $a^{p-1} \not\equiv 1 \pmod{p}$, THEN p is not prime. This conclusion is logically sound. □

11 The Order of an Element and Primitive Roots

Def 11.1: Order of an Element. *Let p be a prime and $a \in \mathbb{Z}$ so that p does not divide a. The order of a modulo p, denoted $\text{ord}_p(a)$, is the minimum positive integer $n \in \mathbb{N} \setminus \{0\}$ so that $a^n \equiv 1 \pmod{p}$.*

Prop 11.1: Order Divisibility Property. *Let p be a prime, $a \in \mathbb{Z}$ so that p does not divide a, and $n \geq 1$. Then $a^n \equiv 1 \pmod{p}$ if and only if $\text{ord}_p(a)$ divides n.*

Proof. We demonstrate both directions of the biconditional.

First Direction ($\Longleftarrow$): Assume $\text{ord}_p(a) \mid n$. $\rightarrow \exists k \in \mathbb{Z}$ so that $n = k \cdot \text{ord}_p(a)$. Then:

$$a^n = a^{k \cdot \text{ord}_p(a)} = (a^{\text{ord}_p(a)})^k$$

By definition of the order, $a^{\text{ord}_p(a)} \equiv 1 \pmod{p}$. Therefore:

$$a^n \equiv (1)^k \equiv 1 \pmod{p}$$

Second Direction ($\Longrightarrow$): Assume $a^n \equiv 1 \pmod{p}$. By the division algorithm, $\exists q, r \in \mathbb{Z}$ so that $n = q \cdot \text{ord}_p(a) + r$, where $0 \leq r < \text{ord}_p(a)$. We evaluate a^n:

$$a^n = a^{q \cdot \text{ord}_p(a) + r} = (a^{\text{ord}_p(a)})^q \cdot a^r$$

Substituting the assumed and known congruences:

$$1 \equiv (1)^q \cdot a^r \pmod{p} \rightarrow a^r \equiv 1 \pmod{p}$$

Since $\text{ord}_p(a)$ is the *minimum* positive integer for which this holds, and $0 \leq r < \text{ord}_p(a)$, the only possibility is that $r = 0$. $\rightarrow \text{ord}_p(a)$ divides n. □

Prop 11.2: Equality of Powers. *Let p be a prime and $g \in (\mathbb{Z}/p\mathbb{Z})^*$. Then $g^x = g^y$ if and only if $x \equiv y \pmod{\text{ord}_p(g)}$.*

Proof. Since $g \in (\mathbb{Z}/p\mathbb{Z})^*$, g is invertible.

$$g^x = g^y \iff g^x \cdot g^{-y} = 1 \iff g^{x-y} = 1$$

Applying the proposition from the previous section (2.1), $g^n = 1$ if and only if $\text{ord}_p(g) \mid n$. Let $n = x - y$. We conclude:

$$g^{x-y} = 1 \iff \text{ord}_p(g) \mid (x - y)$$

By the definition of modular congruence, this is equivalent to: $x \equiv y \pmod{\text{ord}_p(g)}$. □

Def 11.2: Primitive Root. *An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if its powers generate all the elements of $(\mathbb{Z}/p\mathbb{Z})^*$. That is, the set $\{g^1, g^2, \dots, g^{p-1}\}$ is equal to the set $\{1, 2, \dots, p-1\}$.*

Th 11.1: Primitive Root Condition. *An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if and only if $ord_p(g) = p - 1$.*

Proof Outline. For g to be a primitive root, the $p-1$ powers $\{g^1, g^2, \dots, g^{p-1}\}$ must all be distinct. By the Proposition on the Condition for Equality of Powers, $g^i = g^j$ if and only if $i \equiv j \pmod{\text{ord}_p(g)}$. For the powers to be distinct for all $1 \le i < j \le p-1$, we must ensure that $g^{j-i} \neq 1$ for any exponent $j - i$ where $1 \le j - i < p - 1$. This means that the smallest positive exponent k so that $g^k = 1$ must be $p - 1$. By definition, $\text{ord}_p(g) = k$. Therefore, g is a primitive root if and only if $\text{ord}_p(g) = p - 1$. □

Def 11.3. *Let p be a prime, and $a \in \mathbb{Z}$ such that $p \nmid a$. The **order** $ord_p(a)$ of a modulo p is the minimum $n \in \mathbb{N}$ such that $n \ge 1$ and $a^n \equiv 1 \pmod{p}$.*

Prop 11.3. *Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$ and $n \ge 1$. Then $a^n \equiv 1 \pmod{p} \iff ord_p(a) \mid n$. In particular, $ord_p(a) \mid p - 1$ (Fermat's Little Theorem).*

Proof. ($\Rightarrow$) Assume $a^n \equiv 1 \pmod{p}$. Dividing n by $ord_p(a)$, we have $n = q \cdot ord_p(a) + r$, where $0 \le r < ord_p(a)$. Now, $1 \equiv a^n \equiv a^{q \cdot ord_p(a) + r} \equiv a^{q \cdot ord_p(a)} \cdot a^r \equiv (a^{ord_p(a)})^q \cdot a^r \equiv 1^q \cdot a^r \equiv a^r \pmod{p}$. So, $a^r \equiv 1 \pmod{p}$. Since $ord_p(a)$ is the minimum positive exponent such that $a^n \equiv 1 \pmod{p}$, and $r < ord_p(a)$, it must be that $r = 0$. Thus, $n = q \cdot ord_p(a)$, so $ord_p(a) \mid n$.

($\Leftarrow$) If $n = q \cdot ord_p(a)$ for some $q \in \mathbb{Z}$, then $a^n = a^{q \cdot ord_p(a)} = (a^{ord_p(a)})^q \equiv 1^q \equiv 1 \pmod{p}$. □

Cor 11.1. *Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$. $ord_p(a) = \min\{n \in \mathbb{N} : n \ge 1$ so that $a^n \equiv 1 \pmod{p}\} = \min\{n \in \mathbb{N} : n \ge 1$ so that $a^n \equiv 1 \pmod{p}$ and $n \mid p - 1\}$.*

Prop 11.4. *Let p be a prime. If $g \in \mathbb{F}_p^*$, then $g^x = g^y \iff y \equiv x \pmod{ord_p(g)}$.*

Proof. ($\Leftarrow$) Assume $x \equiv y \pmod{ord_p(g)}$. Then $x - y = k \cdot ord_p(g)$ for some integer k. $g^x = g^{y + k \cdot ord_p(g)} = g^y \cdot (g^{ord_p(g)})^k \equiv g^y \cdot 1^k \equiv g^y \pmod{p}$.

($\Rightarrow$) Assume $g^x = g^y$. Without loss of generality, assume $x \ge y$. Since $g \in \mathbb{F}_p^*$ is invertible, multiplying by $(g^{-1})^y$ gives $g^{x-y} = 1$. By the previous Proposition, $ord_p(g) \mid x - y$, which means $x \equiv y \pmod{ord_p(g)}$. □

Def 11.4. *Let p be a prime. A **primitive root** of $\mathbb{F}_p$ is any $g \in \mathbb{F}_p^*$ such that the powers of g generate all the elements of $\mathbb{F}_p^*$, i.e., $\mathbb{F}_p^* = \{1, g, g^2, \dots, g^{p-2}\}$.*

Th 11.2: Characterization of the Primitive Roots. *Let p be a prime. $g \in \mathbb{F}_p^*$ is a primitive root if and only if $ord_p(g) = p-1$.*

Proof. ($\Rightarrow$) Assume g is a primitive root. By definition, $\mathbb{F}_p^* = \{1, g, \dots, g^{p-2}\}$, which means the elements $1, g, \dots, g^{p-2}$ are distinct, and there are $p-1$ of them. If $ord_p(g) = m < p-1$, then $g^m \equiv 1 \pmod p$. This implies that the sequence of powers is periodic with period $m < p-1$, so $g^m = 1$ and $g^{p-1} = 1$ are not the first time the power is 1, and the set $\{1, g, \dots, g^{p-2}\}$ would contain at most m distinct elements, so its cardinality is $|\{1, g, \dots, g^{p-2}\}| \leq m < p-1 = |\mathbb{F}_p^*|$, a contradiction. Thus, $ord_p(g) = p-1$.

($\Leftarrow$) Suppose that $ord_p(g) = p-1$. Claim: The powers $1, g, g^2, \dots, g^{p-2}$ are distinct. If not, there exist $i, j \in \mathbb{Z}$ such that $0 \leq i < j \leq p-2$ and $g^i = g^j$ in $\mathbb{F}_p$. Since g is invertible, $g^{j-i} \equiv 1 \pmod p$. However, $0 < j-i < p-1$. Since $ord_p(g) = p-1$ is the minimum positive exponent such that $g^k \equiv 1 \pmod p$, this is a contradiction. Thus, the $p-1$ elements $\{1, g, \dots, g^{p-2}\}$ are distinct in $\mathbb{F}_p^*$. Since $|\mathbb{F}_p^*| = p-1$, it follows that $\{1, g, \dots, g^{p-2}\} = \mathbb{F}_p^*$, so g is a primitive root. $\square$

Th 11.3: Existence of Primitive Roots. *Let p be a prime. $\mathbb{F}_p$ has at least one primitive root. (Admitted)*

Rem. *Let $p > 2$ be a prime and g be a primitive root in $\mathbb{F}_p$. Then $g^{\frac{p-1}{2}} = -1 \pmod p$.*

Proof. Set $x = g^{\frac{p-1}{2}}$. Then $x^2 = (g^{\frac{p-1}{2}})^2 = g^{p-1} \equiv 1 \pmod p$. The solutions to $x^2 \equiv 1 \pmod p$ in $\mathbb{F}_p$ are $x = 1$ or $x = -1$. If $x = 1$, then $g^{\frac{p-1}{2}} \equiv 1 \pmod p$. This would imply that $ord_p(g) \mid \frac{p-1}{2}$. However, $ord_p(g) = p-1$ because g is a primitive root, and $p-1 \nmid \frac{p-1}{2}$. Thus, $x \neq 1$, so $x = -1$, meaning $g^{\frac{p-1}{2}} \equiv -1 \pmod p$. $\square$

Rem. *Let $p > 2$ be a prime. If g is a square in $\mathbb{F}_p$, then g is not a primitive root in $\mathbb{F}_p$.*

Proof. Let g be a square, so $g = h^2$ for some $h \in \mathbb{F}_p^*$. Then $g^{(p-1)/2} = (h^2)^{(p-1)/2} = h^{p-1} \equiv 1 \pmod p$ by Fermat's Little Theorem. Since $g^{(p-1)/2} \equiv 1 \pmod p$, by the proposition on order, $ord_p(g) \mid \frac{p-1}{2}$. Since $\frac{p-1}{2} < p-1$ (as $p > 2$), we have $ord_p(g) < p-1$. By the characterization theorem, g is not a primitive root. $\square$

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lemma 11.1: Effective Test for a Primitive Root Candidate. *Let $p > 2$ be prime and $p - 1 = \prod_{i=1}^m p_i^{a_i}$ be the prime factorization of $p-1$, with p_i distinct primes and $a_i \geq 1$ for all $i = 1, \dots, m$. Then $g \in \mathbb{F}_p^*$ is a primitive root if and only if $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all $i = 1, \dots, m$.*

Proof. ($\Rightarrow$) If g is a primitive root, then $ord_p(g) = p-1$. If $g^{(p-1)/p_i} \equiv 1 \pmod p$ for some i, then $ord_p(g) \mid \frac{p-1}{p_i}$. This means $p-1 \mid \frac{p-1}{p_i}$, which is only possible if $p_i = 1$, but p_i is a prime, a contradiction. Thus, $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all i.

($\Leftarrow$) Assume $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all $i = 1, \dots, m$. Let $m = ord_p(g)$. We know $m \mid p-1$. We can write m as $m = p_1^{f_1} \cdot \dots \cdot p_m^{f_m}$ where $0 \leq f_i \leq a_i$ for all i. Suppose $m < p-1$. Then there must be at least one index i such that $f_i < a_i$. For this index i, we have $m \mid \frac{p-1}{p_i}$. Since $m \mid \frac{p-1}{p_i}$, by the proposition on order, we must have $g^{(p-1)/p_i} \equiv 1 \pmod p$, which contradicts our assumption. Therefore, it must be that $m = p - 1$, so $ord_p(g) = p - 1$, and g is a primitive root. □

Alg 11.1: Finding a Primitive Root. *Let $p > 2$ be prime and $p - 1 = \prod_{i=1}^{m} p_i^{a_i}$ be the prime factorization of $p - 1$, with p_i distinct primes and $a_i \geq 1$ for $i = 1, \ldots, m$.*

1. *Start with $g = 2$.*
2. *If $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ for all $i = 1, \ldots, m$, then g is a primitive root.*
3. *If not, increment $g \to g + 1$ and return to Step 2.*

12 Number of Primitive Roots

Th 12.1: Characterization of Powers as Primitive Roots. *Let p be a prime and g be a primitive root of $\mathbb{F}_p$. For any $i \in \{1, \dots, p-2\}$, we have:*

1. *g^i is a primitive root of $\mathbb{F}_p \iff \gcd(i, p-1) = 1$.*
2. *$\mathbb{F}_p$ has exactly $\phi(p-1)$ primitive roots.*

Proof of 1. Let $p - 1 = p_1^{a_1} \cdots p_m^{a_m}$ with p_j distinct primes and $a_j \geq 1$. By the effective test for a primitive root (Lemma from Lesson 5), g^i is not a primitive root $\iff (g^i)^{(p-1)/p_j} \equiv 1 \pmod{p}$ for some j. We analyze the condition $(g^i)^{(p-1)/p_j} \equiv 1 \pmod{p}$:

$$(g^i)^{(p-1)/p_j} \equiv 1 \pmod{p} \iff ord_p(g) \mid \frac{i(p-1)}{p_j} \quad \text{since } g \text{ is a primitive root, } ord_p(g) = p-1$$

$$\iff p - 1 \mid \frac{i(p-1)}{p_j}$$

$$\iff p_j \mid i$$

Thus, g^i is not a primitive root $\iff p_j \mid i$ for some j. Since the p_j are the distinct prime factors of $p-1$, the condition "$p_j \mid i$ for some j" is equivalent to $\gcd(i, p-1) > 1$. Therefore, g^i is a primitive root $\iff \gcd(i, p-1) = 1$. □

Proof of 2. Since g is a primitive root, the elements of $\mathbb{F}_p^*$ are given by the distinct powers $\{1, g, \dots, g^{p-2}\}$. From part 1, the other primitive roots are the powers g^i such that $1 \leq i \leq p-2$ and $\gcd(i, p-1) = 1$. The number of such integers i in the range $\{1, \dots, p-2\}$ (or $\{1, \dots, p-1\}$) that are relatively prime to $p-1$ is exactly $\phi(p-1)$, where ϕ is Euler's totient function. Therefore, $\mathbb{F}_p$ has $\phi(p-1)$ primitive roots. □

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lemma 12.1. *Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. Then $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$.*

Proof. Let $d = \gcd(a, b)$. We set $a = a'd$ and $b = b'd$, where $\gcd(a', b') = 1$. The statement $a \mid bc$ is equivalent to $a'd \mid b'dc$. Since $d \neq 0$, we can cancel d: $a' \mid b'c$. Because $\gcd(a', b') = 1$, by Euclid's Lemma, $a' \mid b'c \iff a' \mid c$. Substituting back $a' = \frac{a}{d} = \frac{a}{\gcd(a,b)}$, we get the equivalence: $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$. □

Prop 12.1: The Order of a Power. *Let p be a prime, g be a primitive root in $\mathbb{F}_p$, and $i \in \{1, \dots, p-2\}$. Then $ord_p(g^i) = \frac{p-1}{\gcd(p-1,i)}$.*

Proof. The order $ord_p(g^i)$ is defined as $m = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (g^i)^n \equiv 1 \pmod{p}\}$. The congruence $(g^i)^n \equiv 1 \pmod{p}$, or $g^{in} \equiv 1 \pmod{p}$, is equivalent to $ord_p(g) \mid in$. Since g is a primitive root, $ord_p(g) = p - 1$. Thus, we are looking for:

$$ord_p(g^i) = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (p-1) \mid in\}$$

Using the preceding Lemma with $a = p - 1$, $b = i$, and $c = n$, the condition $(p-1) \mid in$ is equivalent to $\frac{p-1}{\gcd(p-1,i)} \mid n$. The minimum positive integer n such that $\frac{p-1}{\gcd(p-1,i)} \mid n$ is $n = \frac{p-1}{\gcd(p-1,i)}$. Therefore, $ord_p(g^i) = \frac{p-1}{\gcd(p-1,i)}$. □

Cor 12.1: Alternative Proof. *Let g be a primitive root in $\mathbb{F}_p$. Then g^i is a primitive root $\iff \gcd(p-1, i) = 1$.*

Proof. By the Characterization Theorem, g^i is a primitive root $\iff ord_p(g^i) = p - 1$. Using the Proposition on the order of a power:

$$ord_p(g^i) = p - 1 \iff \frac{p-1}{\gcd(p-1, i)} = p - 1$$

$$\iff \gcd(p-1, i) = 1$$

□

Artin's Conjecture

Rem. *If $g \in \mathbb{N}$ is a perfect square, there is no prime p such that $p > g$ and g is a primitive root in $\mathbb{F}_p$.*

Th 12.2: Artin's Conjecture (Hooley, 1967 - under GRH). *If $g \geq 2$ is not a perfect square, the set of primes p for which g is a primitive root in $\mathbb{F}_p$ has a positive density.*

$$\lim_{x \to +\infty} \frac{\#\{p \leq x : p \text{ is prime and } g \text{ is a primitive root in } \mathbb{F}_p\}}{\#\{p : p \text{ is prime, } p \leq x\}} = C_g > 0$$

For $g = 2$, the constant is $C_2 \approx 0.3739558\ldots$

Th 12.3: Roger-Heath-Brown, 1985 - unconditional. *There are at most two primes that are not primitive roots for infinitely many primes.*
Rem. *This theorem implies that among the first few primes, like 2, 3, and 5, at least one of them must be a primitive root for infinitely many primes.*